

Sydney Girls High School



November 2015

MATHEMATICS EXTENSION 1

Year 12

ASSESSMENT TASK 1 FOR HSC 2016

Time Allowed: 60 minutes + 5 minutes reading time

Topics: Integration, Trigonometric Functions II, Parametric Equations

General Instructions:

- There are Seven (7) questions each worth 8 marks
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Mathematics reference sheet is provided.

TOTAL: 56 MARKS

Student's Name:Teacher Name:

QUESTION 1**Start a new page****Marks**

(a) Find $\int \frac{dx}{(2x-1)^4}$ 2

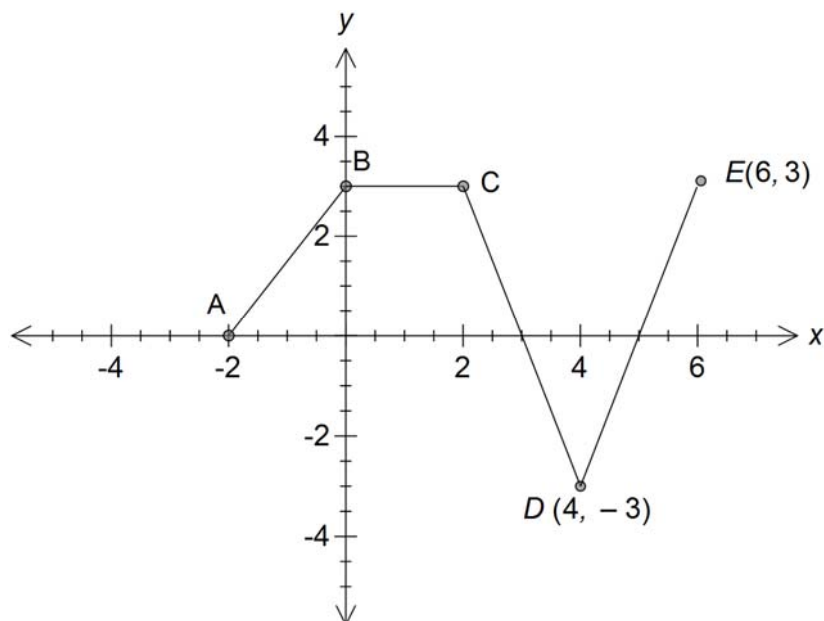
(b) Evaluate $\int_{-2}^3 (5-x^2) \, dx$ 2

(c) Find the exact value of $\cos 75^\circ$. 2

(d) Find the Cartesian equation of a function whose parametric equations are $x = 2t$ and $y = t^2 - 1$. 2

- (a) Given that $\int_1^2 f(x) \, dx = 3$ and $f(x) + f(-x) = 0$ find $\int_{-2}^{-1} f(x) \, dx$. 1
- (b) Find the area bounded by the curve $y = 9 - x^2$ and the x axis. 2
- (c) Given that the exact value of $\sin x$ is $\frac{7}{15}$ and the exact value of $\cos y$ is $\frac{1}{\sqrt{3}}$ find the exact value of $\sin (x + y)$ where x and y are acute. 3
- (d) The point $(8t, 4t^2)$ lies on the parabola $x^2 = 16y$. Find the equation of the chord whose endpoints have parameters 2 and -1. 2

(a)



Evaluate

(i) $\int_0^2 f(x) \, dx$ 1

(ii) $\int_3^5 f(x) \, dx$ 1

(iii) $\int_{-2}^6 f(x) \, dx$ 1

(b) Solve $\sin 2x - \sin x = 0$ for $0 \leq x \leq 2\pi$. 3

(c) For the parabola $x = 6t, y = 3t^2$ find the equation of the normal at $t = 2$. 2

Question 4 **Start a new page**

Marks

- (a) Evaluate $\int_0^2 3^x dx$ using Simpson's rule with 5 function values. 2

Answer correct to 2 decimal places.

- (b) (i) Express $3\cos 2x - 4\sin 2x$ in the form $R\cos(2x + \alpha)$. 2

- (ii) Hence solve $6\cos 2x - 8\sin 2x = 2$ for $0^\circ \leq x \leq 360^\circ$. 2

- (c) From an external point two tangents are drawn to the parabola $y = x^2$.
Find the co-ordinates of the external point if the equation of the chord of contact of the two tangents is $y = 4x + 3$. 2

(a) The region bounded by $y = 3$, $x = y^2$ and the y axis is rotated around the x axis.

(i) Sketch the region above on a number plane. 1

(ii) Find the exact volume of the solid of rotation. 2

(b) Prove that $\tan \frac{x}{2} = \frac{\sin x - \cos x + 1}{\sin x + \cos x + 1}$ 2

(c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$

(i) Show that $m_{PQ} = \frac{p+q}{2}$. 1

(ii) If $pq = -1$ show that PQ is a focal chord. 2

Question 6 Start a new page

Marks

(a) (i) Find the points of intersection of $y = x^2 - 3x + 1$ and $y = 2x - 5$. 2

(ii) Find the area bounded by $y = x^2 - 3x + 1$ and $y = 2x - 5$. 2

(b) Solve $3\sin\theta + 2\cos\theta = 1$ by using the t method for $0^\circ \leq \theta \leq 360^\circ$. 2

(c) Find the equation of the locus of the midpoint of PQ where P is the point $(2ap, ap^2)$ and Q is the point $(6ap, 9ap^2)$ on the parabola $x^2 = 4ay$. 2

(a) Find the area bounded by the curve $y = 9x - x^3$ and the x axis. 2

(b) Find the exact value of $\tan \frac{3\pi}{8}$. 3

(c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. 3
The tangents at P and Q meet at R , and R lies on the parabola $x^2 = -4ay$.
Show that $p^2 + q^2 + 6pq = 0$.

Question 1 - Maths Ext 1 - Assessment Task 1 Year 12

(a) $\int \frac{dx}{(2x-1)^4}$ $\int (2x-1)^{-4} dx$ $\frac{(2x-1)^{-3}}{-3 \times 2} + C$ $= -\frac{(2x-1)^{-3}}{6} + C$ $= -\frac{1}{6(2x-1)^3} + C$ (a) need to be careful adding a one to a negative number (2)	(b) $\int_{-2}^3 (5-x^2) dx$ $\left[5x - \frac{x^3}{3} \right]_{-2}^3$ $\left[5(3) - \frac{(3)^3}{3} \right] - \left[5(-2) - \frac{(-2)^3}{3} \right]$ $[15-9] - [-10+\frac{8}{3}]$ $= \frac{40}{3}$ (2)
(c) $\cos 75$ $\cos(30+45)$ $\cos 30 \times \cos 45 - \sin 30 \sin 45$ $= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$ $= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$ $= \frac{\sqrt{3}-1}{2\sqrt{2}}$ or $\frac{\sqrt{6}-\sqrt{2}}{4}$ (2)	(d) $x=2t$ $y=t^2-1$ $t = \frac{x}{2}$ $t^2 = \frac{x^2}{4}$ $\frac{x^2}{4} = y+1$ $x^2 = 4(y+1)$ (2)

(c) Few students did not use correct addition of angles formula.

Q₂

$$a) \int_1^2 f(x) dx = 3 \quad ; \quad f(x) + f(-x) = 0$$
$$f(x) = -f(-x)$$

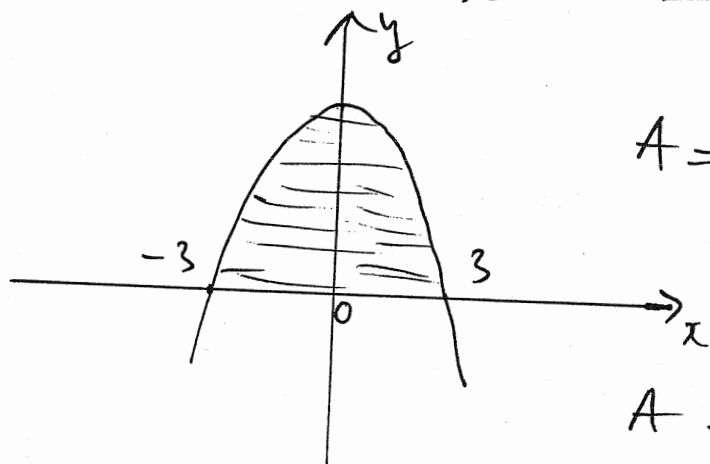
$$\int_{-2}^{-1} f(x) dx = - \int_{-1}^{-2} f(x) dx$$

$$= - \int_1^2 f(-x) dx = -3 \checkmark$$

Some students could not recognise this is an odd function.

$$b) y = 9 - x^2$$

$$x\text{-intercepts} : (3-x)(3+x) = 0$$
$$x = \pm 3$$



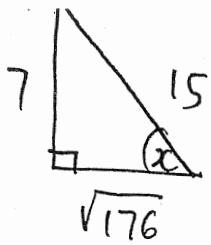
$$A = \int_{-3}^3 (9 - x^2) dx$$

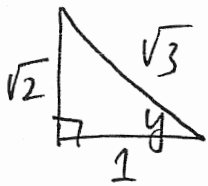
$$A = 2 \int_0^3 (9 - x^2) dx$$

$$A = 2 \left[9x - \frac{x^3}{3} \right]_0^3$$

$$A = 2 \left[9 \times 3 - \frac{3^3}{3} - 0 \right]$$

$$A = 36 \text{ units}^2$$

c) $\sin x = \frac{7}{15}$  $\cos x = \frac{\sqrt{176}}{15}$ ✓

$\cos y = \frac{1}{\sqrt{3}}$  $\sin y = \frac{\sqrt{2}}{\sqrt{3}}$

$$\begin{aligned}\sin(x+y) &= \sin x \cdot \cos y + \cos x \sin y \\ &= \frac{7}{\sqrt{15}} \cdot \frac{1}{\sqrt{3}} + \frac{\sqrt{176}}{15} \cdot \frac{\sqrt{2}}{\sqrt{3}} \quad \checkmark \\ &= \frac{7 + \sqrt{352}}{15\sqrt{3}} = \frac{7\sqrt{3} + 4\sqrt{66}}{45} \quad \checkmark\end{aligned}$$

* Some students could not find the ratios $\cos x$ and $\sin y$, hence could not evaluate $\sin(x+y)$.

d) At $A(8t, 4t^2)$

At $t = 2 \therefore P(16, 16)$

$t = -1 \therefore Q(-8, 4)$

Eq of the chord PQ

$y - y_1 = m(x - x_1)$

$y - 16 = \frac{4 - 16}{-8 - 16} (x - 16)$

$x - 2y + 16 = 0$

Alternate method

$y = \frac{p+q}{2}x - a_1$

$y = \frac{2+(-1)}{2}x - 4(2)(-1)$

$y = \frac{1}{2}x + 8$

$x - 2y + 16 = 0$

Ext 1 2015 Assessment 1

3a) i)

$$\int_0^2 f(x) dx = 2 \times 3 = 6$$

✓

Many students calculated the area instead of Evaluating the Integral.

ii) $\int_3^5 f(x) = -\frac{1}{2} \times 2 \times 3 = -3$

✓

iii) $\int_{-2}^6 f(x) dx = \frac{1}{2} \times 2 \times 3 + 6 + \frac{1}{2} \times 3 \times 1 - 3 + \frac{1}{2} \times 3 \times 1 = 9$

✓

b) $2 \sin x \cos x - \sin x = 0$

$$\sin x (2 \cos x - 1) = 0$$

✓

$$\sin x = 0$$

$$x = 0, \pi, 2\pi$$

Some students didn't have all the solutions

$$\cos x = \frac{1}{2}$$

✓

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

✓

c) $t = 2$

$$\therefore x = 12, y = 12$$

This question was done well

$$m_1 = 2 \quad m_2 = -\frac{1}{2}$$

✓ ✓

$$y - 12 = -\frac{1}{2}(x - 12)$$

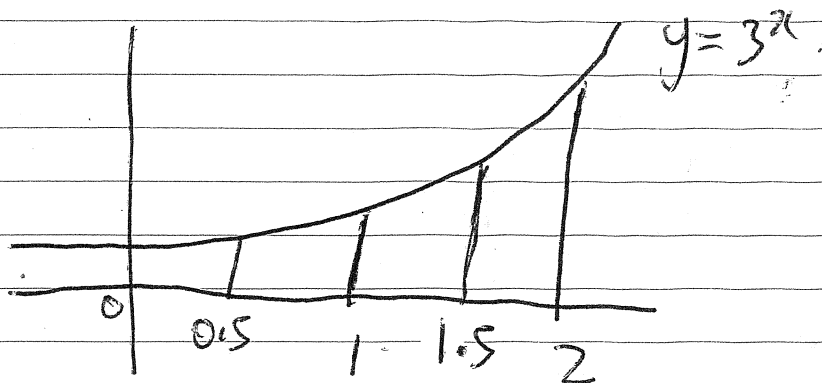
$$2y - 24 = -x + 12$$

$$\boxed{x + 2y - 36 = 0}$$

ASSESSMENT TASK 1 2015 Ext 1.

Q4.

(a)



$$\int_a^b f(x) dx = \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)].$$

$$\int_0^1 3^x dx + \int_1^2 3^x dx$$

$$= \frac{1}{6} [3^0 + 4(3^{0.5}) + 3^1] + \frac{1}{6} [3^1 + 4(3^{1.5}) + 3^2]$$

$$= \frac{1}{6} (4 + 4\sqrt{3}) + \frac{1}{6} (12 + 12\sqrt{3})$$

$$= \frac{1}{6} (16 + 16\sqrt{3})$$

$$= \frac{8}{3} + \frac{8\sqrt{3}}{3} \approx 7.29$$

If you can't remember Simpson's rule, it's on the reference sheet.

$$(b) (i) R \cos(2x + \alpha) = R \cos \alpha \cos 2x - R \sin \alpha \sin 2x$$

$$3 \cos 2x - 4 \sin 2x$$

$$R \sin \alpha = 4$$

$$\tan \alpha = \frac{4}{3}$$

$$R \cos \alpha = 3$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\approx 53^\circ 8'$$

$$R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 16 + 9$$

$$R = 5$$

$$5 \cos(2x + 53^\circ 8')$$

$$(ii) 6 \cos 2x - 8 \sin 2x = 2$$

$$3 \cos 2x - 4 \sin 2x = 1$$

$$5 \cos(2x + \alpha) = 1$$

$$\text{Let } \beta = \cos^{-1}\left(\frac{1}{5}\right)$$

$$2x + \alpha = \beta, 360^\circ - \beta, 360^\circ + \beta, 720^\circ - \beta$$

$$2x = \beta - \alpha, 360^\circ - \beta - \alpha, 360^\circ + \beta - \alpha, 720^\circ - \beta - \alpha$$

$$x = \frac{\beta - \alpha}{2}, \frac{360^\circ - \beta - \alpha}{2}, \frac{360^\circ + \beta - \alpha}{2}, \frac{720^\circ - \beta - \alpha}{2}$$

$$= 12^\circ 40', 114^\circ 12', 192^\circ 40', 294^\circ 12'$$

Some students only found solutions

$$0^\circ \leq x \leq 180^\circ.$$

(c) $y = x^2$

$$4\left(\frac{1}{4}\right)y = x^2$$

$$a = \frac{1}{4}$$

Chord of Contact

$$x_0 x = 2a(y + y_0)$$

$$x_0 x = \frac{1}{2}(y + y_0)$$

Given $y = 4x + 3$

$$4x = y - 3$$

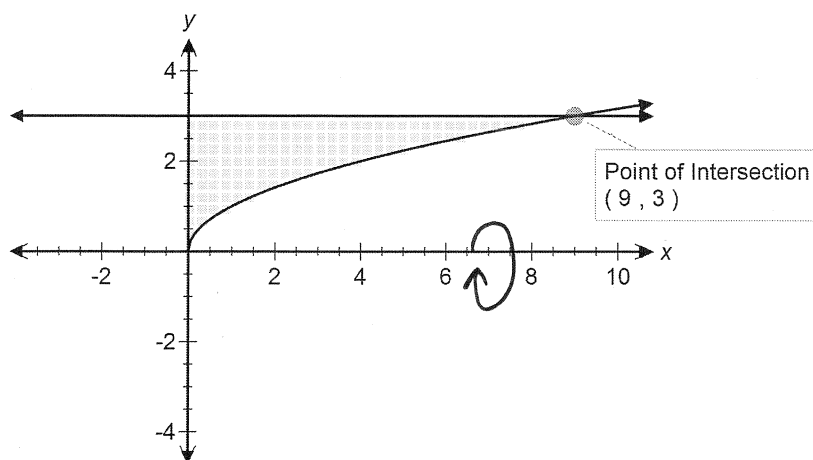
$$2x = \frac{1}{2}(y - 3)$$

Ext. Point $(2, -3)$

Some student tried to solve the two equations simultaneously. While it is possible to do this question that way it proved to be too difficult.

Question 5 (8 Marks)

a) i)



$$\begin{aligned}
 \text{ii) } V &= \pi r^2 h - \pi \int y^2 dy \\
 &= \pi \times 3^2 \times 9 - \pi \int_0^9 x \, dx \\
 &= 81\pi - \pi \left[\frac{x^2}{2} \right]_0^9 \\
 &= 81\pi - \frac{81\pi}{2} \\
 &= 40\frac{1}{2}\pi \text{ units}^3.
 \end{aligned}$$

OR

$$\begin{aligned}
 V &= \pi \int [f(x)^2 - g(x)^2] \, dx. \\
 &= \pi \int_0^9 [(3)^2 - y^2] \, dx \\
 &= \pi \int_0^9 (9 - x) \, dx \\
 &= \pi \left[9x - \frac{x^2}{2} \right]_0^9 \\
 &= \pi \left[81 - \frac{81}{2} \right] \\
 &= \frac{81}{2}\pi \text{ units}^3.
 \end{aligned}$$

Many students could not work out the point of intersection which then determined the limits of integration.

MOST students lost a mark because of the incorrect method of finding the volume of the shaded region.

$$b) \text{ RHS} = \frac{\sin x - \cos x + 1}{\sin x + \cos x + 1}$$

$$= \left(\frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} + 1 \right) \div \left(\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} + 1 \right)$$

$$= \left(\frac{2t - 1 + t^2 + 1 + t^2}{1+t^2} \right) \div \left(\frac{2t + 1 - t^2 + 1 + t^2}{1+t^2} \right)$$

(2)

$$= \left(\frac{2t^2 + 2t}{1+t^2} \right) \times \left(\frac{1+t^2}{2t+2} \right)$$

$$= \frac{2t(t+1)}{2(t+1)}$$

$$= t$$

$$= \text{LHS.}$$

Many students could not complete this proof and lost one mark.

OR. (A few students proved as follows):

$$\text{RHS} = \frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$$

$$= \frac{\sin x + (1 - \cos x)}{\sin x + (-1 + \cos x)}$$

$$= \frac{2 \sin \frac{x}{2} \cos \frac{x}{2} + 2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2} + 2 \cos^2 \frac{x}{2}}$$

$$= \frac{2 \sin \frac{x}{2} \cos \frac{x}{2} + 2 \cos^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2} + 2 \cos^2 \frac{x}{2}}$$

$$= \frac{2 \sin \frac{x}{2} \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)}{2 \cos \frac{x}{2} \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)}$$

$$= \tan \frac{x}{2}$$

$$= t = \text{LHS.}$$

$$\begin{aligned}
 \text{c) i) } m_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\
 &= \frac{a(p-q)(p+q)}{2a(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}
 \tag{1}$$

ii) Equation of PQ:

$$y - ap^2 = \frac{p+q}{2}(x - 2ap)$$

$$2y - 2ap^2 = px - 2ap^2 + qx - 2apq$$

$$y = \frac{(p+q)}{2}x - apq$$

step ① If $pq = -1$
 then: $y = \frac{(p+q)}{2}x + a$

step ② Substitute focus $(0, a)$: ②

$$\begin{array}{ll}
 \text{LHS} = y & \text{RHS} = \left(\frac{p+q}{2}\right) \times 0 + a \\
 = a & = a
 \end{array}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ focus satisfies eqn of chord PQ
 if $pq = -1$.

Students lost one mark if they
 showed step ② before step ①.

$$6 (a) (i) x^2 - 3x + 1 = 2x - 5$$

$$x^2 - 5x + 6 = 0$$

$$(x-3)(x-2) = 0$$

$$x = 3, 2$$

$$\text{When } x=3, y = 2 \times 3 - 5 = 1$$

$$\text{When } x=2, y = 2 \times 2 - 5 = -1$$

When finding points both co-ordinates need to be found.

$$(ii) \int_2^3 (2x - 5 - (x^2 - 3x + 1)) dx$$

$$= \int_2^3 (-x^2 + 5x - 6) dx$$

$$= \left[-\frac{x^3}{3} + \frac{5x^2}{2} - 6x \right]_2^3$$

$$= -9 + \frac{45}{2} - 18 + \frac{8}{3} + 10 + 12$$

$$= \frac{1}{6}$$

$$(b) 3 \times \frac{2t}{1+t^2} + 2 \times \frac{1-t^2}{1+t^2} = 1$$

$$6t + 2 - 2t^2 = 1 + t^2$$

$$0 = 3t^2 - 6t - 1$$

$$t = \frac{6 \pm \sqrt{6^2 - 4 \times 3 \times -1}}{2 \times 3}$$

$$= \frac{6 \pm \sqrt{48}}{6}$$

$$= \frac{6 \pm 4\sqrt{3}}{6}$$

$$= \frac{3 \pm 2\sqrt{3}}{3}$$

$$\tan \frac{G}{2} = 65^\circ, 171^\circ$$

$$\therefore G = 130^\circ, 342^\circ$$

$\frac{G}{2}$ has to be between 0° and 180° .

$$(c) x = \frac{2ap + 6ap}{2} \quad y = ap^2 + 9ap^2$$

$$= 4ap$$

$$p = \frac{x}{4a}$$

$$= \frac{10ap^2}{2}$$

$$= 5ap^2$$

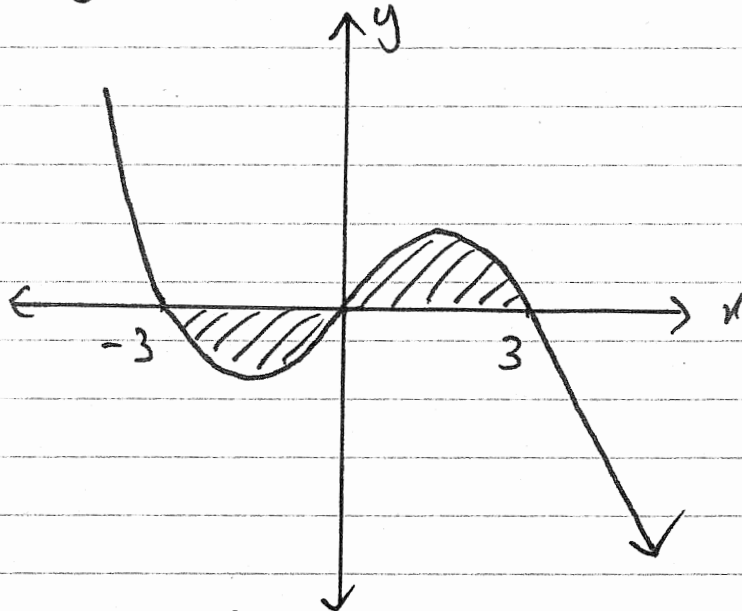
$$= 5a \times \left(\frac{x}{4a} \right)^2$$

$$= \frac{5x^2}{16a}$$

There must be no " p " in the final answer.

QUESTION 7

(a) $y = x(9 - x^2)$



$$A = 2 \int_0^3 (9x - x^3) dx$$

$$= 2 \left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^3$$

$$= 2 \left(\frac{81}{2} - \frac{81}{4} - (0) \right)$$

$$\therefore A = 40.5 \text{ units}^2$$

This question was done well by most students. Some students did not identify the correct intercepts on the graph or didn't identify the correct region of interest.

(b) Consider $t = \tan \frac{\theta}{2}$ where $\theta = \frac{3\pi}{4}$

$$\therefore t = \tan \frac{3\pi}{8}$$

$$\tan \theta = \frac{2t}{1-t^2} \quad \therefore \tan \frac{3\pi}{4} = \frac{2t}{1-t^2}$$

$$-1 = \frac{2t}{1-t^2}$$

$$-1 + t^2 = 2t$$

$$t^2 - 2t - 1 = 0$$

$$(t-1)^2 = 2$$

Many students failed to make the connection to the t-formula on this question.

(b) Continued

To get this question out fully, you need to identify that there is only one answer and eliminate the minus sign.

$$\therefore t = 1 \pm \sqrt{2}$$

$$\tan \frac{3\pi}{8} > 0 \quad \left(\frac{3\pi}{8} \text{ in quadrant 1} \right)$$

$$\therefore \tan \frac{3\pi}{8} = 1 + \sqrt{2}$$

(c) Tangent at $P(2ap, ap^2)$

$$m_T = p$$

$$\therefore y - ap^2 = p(x - 2ap)$$

$$y = px - ap^2$$

$$\text{Tangent at } Q: y = qx - aq^2$$

R:

$$px - ap^2 = qx - aq^2$$

$$x(p - q) = a(p^2 - q^2)$$

$$= a(p - q)(p + q)$$

$$\therefore x = a(p + q)$$

$$y = p \cdot a(p + q) - ap^2$$

$$= ap^2 + apq - ap^2 = apq$$

$$\therefore R \text{ is } (a(p + q), apq)$$

Since R lies on $x^2 = -4ay$

$$(a(p + q))^2 = -4a(apq)$$

$$a^2(p^2 + 2pq + q^2) = -4a^2(pq)$$

$$\therefore p^2 + 2pq + q^2 = -4pq$$

$$\text{i.e. } p^2 + 6pq + q^2 = 0$$

Finding the coordinates of R should have been fairly straight forward as a parametrics question though it seems that some students may have struggled with time at the end of the paper.

Proving the required result was difficult for students.